TERMS OF REFERENCE (TOR) - TECHNICAL SCOPE

Assignment: LOAN E-060009-001 PAK: Upgradation of Pakistan Railways Main Line-1 (K

1. ASSIGNMENT BACKGROUND & STRATEGIC OBJECTIVES

The Asian Development Bank (ADB), in coordination with the Executing Agency in Pakistan, requires the services of an experienced consulting firm to provide high-level professional advisory, geospatial analytics, and technical implementation for LOAN E-060009-001 PAK: Upgradation of Pakistan Railways Main Line-1 (Karachi-Rohri Section) Project - PR/ML-1/Cons-1 Project Management and Construction Supervision Consultant. The principal objective is to establish robust, sustainable, and climate-resilient spatial frameworks, ensuring strict adherence to ADB Safeguard Policy Statement (SPS) and international best engineering practices.

2. COMPREHENSIVE SCOPE OF WORK & WORK PACKAGES

Work Package 1: Baseline Assessment, Data Collection & Spatial Data Infrastructure (SDI) Review.
Work Package 2: Drone LiDAR / High-Resolution Satellite Data Ingestion, Vector Processing & CAD Conversion.
Work Package 3: Development of Interactive WebGIS Geodatabase, Cloud Analytics & Real-Time Dashboards.
Work Package 4: Institutional Strengthening, Technology Transfer, and Comprehensive User Training.

3. REQUIRED KEY EXPERTS & TEAM COMPOSITION

? Team Leader / Senior Multilateral Project Manager (International, 12 Person-Months): Minimum 15 years in managing ADB/multilateral infrastructure projects.
? Senior Geospatial / GIS Systems Architect (International/National, 10 Person-Months): Master's degree in Geoinformatics with expertise in enterprise spatial geodatabases.
? Remote Sensing & Photogrammetry Specialist (National, 8 Person-Months): Extensive background in sub-meter satellite and drone LiDAR point cloud classification.
? Environmental & Social Safeguards Specialist (National, 6 Person-Months): Specialist in ADB SPS 2009 and environmental management frameworks.
? Full-Stack WebGIS Software Engineer (National, 8 Person-Months): Expert in React, Python, PostgreSQL/PostGIS, and MapLibre/OpenLayers.

4. DELIVERABLES, REPORTING SCHEDULE & MILESTONES

? Inception Report & Detailed Quality Assurance Plan (QAP) ? Month 1
? Baseline Spatial Data Infrastructure & Field Survey Report ? Month 3
? Draft Geospatial Analytics Engine & Interim System Demonstration ? Month 6
? Final Operational WebGIS Geoportal, User Manuals & Source Code Handover ? Month 12